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RESEARCH PAPER

Data-Efficient Generative Segmentation for High-Resolution Breast CT: A Flow-Matching Approach with Physics-Informed Priors

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Abstract

Breast Computed Tomography (BCT) provides isotropic high-resolution three-dimensional breast imaging without compression and supports quantitative tissue characterization using Hounsfield Units (HU). However, automated tumor segmentation in BCT is difficult because each scan contains massive volumetric data while annotated cohorts remain small, and tumor boundaries are physically ambiguous due to partial-volume mixing. In this setting, conventional discriminative segmentation models are prone to overfitting and tend to collapse boundary ambiguity into overconfident predictions. We therefore propose a physics-informed generative framework that formulates BCT segmentation as distribution learning with Flow-Matching. The method combines three components: HU-based tumor-candidate localization to reduce background dominance, region-of-interest-centered 3D patch augmentation and random affine perturbations, and a lightweight 3D UNet that predicts latent velocity fields in a variational autoencoder space and generates masks through Ordinary Differential Equation (ODE) integration. The framework is evaluated with five-fold cross-validation on a clinical BCT cohort. Compared with a deterministic UNet baseline, the proposed approach improves mean candidate-region Dice from 0.40 to 0.44 and reduces false positives per scan from 3.0 to 1.5, while maintaining comparable lesion-level detection sensitivity (45.1% versus 44.7%). Importantly, the residual performance gap is concentrated in small, low-contrast lesions and is governed primarily by lesion-level learning and detection rather than by contour refinement, while deterministic ODE inference remains computationally tractable for high-resolution deployment. These results support Flow-Matching as a data-efficient strategy for screening-oriented BCT tumor analysis and indicate that future gains will depend primarily on more reliable tumor localization and higher lesion-level detection sensitivity, with boundary-level refinement and false-positive burden as secondary objectives.

1 Introduction

Breast cancer is now the most commonly diagnosed malignancy worldwide, and imaging remains the cornerstone of early detection Sung et al. [2021]. However, conventional mammography projects complex three-dimensional anatomy onto a two-dimensional plane, making tumor conspicuity strongly dependent on tissue overlap and breast density Boyd et al. [2007]. Digital Breast Tomosynthesis partially mitigates tissue superposition, but as a limited-angle technique it still provides anisotropic depth information and incomplete volumetric recovery in complex dense tissue patterns. Dedicated breast Computed Tomography systems address this geometric limitation by providing true volumetric imaging with isotropic high-resolution voxels and no breast compression Prionas et al. [2010], Weber et al. [2024], Ruby et al. [2020]. Spiral Breast Computed Tomography (SBCT) with photon-counting detector technology represents the most recent iteration of this approach. Equipped with photon-counting detector technology, dedicated BCT systems can reach isotropic voxel resolutions on the order of 150 μm, improving conspicuity for subtle structural features while maintaining quantitative attenuation measurements in Hounsfield Units (HU) Weber et al. [2024], Wieler et al. [2021], Landsmann et al. [2022]. Recent non-contrast BCT evidence shows that malignant tumors are substantially hyperdense relative to dense glandular tissue (approximately 60–62 HU versus 28–33 HU), with significant separability and strong discrimination performance Weber et al. [2024]. These findings establish a physics-informed signal basis for automated tumor localization in dense breasts. At the same time, exploiting this signal in practice requires algorithms that can process massive high-resolution volumes under limited annotated data availability in this emerging modality Shim et al. [2022].

The central methodological barrier in BCT segmentation is the mismatch between data scale and supervision scale. A typical BCT acquisition contains on the order of 10^9 voxels, while clinically curated annotated cohorts in this emerging modality frequently remain low Shim et al. [2022], Caballo et al. [2020]. Prior BCT and breast-CT tumor-segmentation studies have demonstrated feasibility, but most pipelines remain fundamentally discriminative and predominantly 2D-oriented in their supervision strategy Caballo et al. [2020], Shim et al. [2022], Caballo et al. [2021, 2018]. Shim *et al.* developed fully automated breast tissue segmentation workflows on BCT volumes using seeded watershed and region growing algorithms Shim et al. [2022], while Caballo *et al.* demonstrated that deep-learning mass segmentation in dedicated breast CT can yield radiomic features statistically equivalent to expert contours Caballo et al. [2020], and further adopted multi-view 2D patch-based representations combining handcrafted and CNN features for mass classification Caballo et al. [2021]. Earlier unsupervised approaches established tissue classification baselines for breast CT dosimetry Caballo et al. [2018]. These approaches establish important baselines, yet they still enforce single-mask supervision for physically ambiguous boundaries, which can weaken lesion-level detection robustness in sparse-data BCT settings.

Within such volumes, sub-centimeter tumors can occupy less than 0.01% of all voxels, producing extreme foreground-background imbalance and weak tumor gradients during optimization. In this regime, discriminative frameworks such as nnU-Net, although highly successful in larger biomedical benchmarks, are vulnerable to overfitting and unstable calibration when supervision is sparse relative to dimensionality Isensee et al. [2021]. Standard volumetric augmentation can partially improve robustness, but under very small cohorts its effective diversity remains constrained by the narrow empirical training distribution, even when extensive geometric and intensity transforms are applied Isensee et al. [2021], Pérez-García et al. [2021]. More fundamentally, deterministic segmentation pipelines collapse a potentially multimodal posterior into a single binary target and therefore assume that expert delineation is uniquely correct Schmidt et al. [2023]. That assumption is violated in BCT because boundary voxels are physically mixed by partial volume effects, yielding intermediate attenuation values that are not unambiguously tumor or parenchyma Caballo et al. [2021], Weber et al. [2024]. Consistent with this physics-driven ambiguity, inter-reader agreement in breast CT tumor delineation is known to be limited, indicating that no single contour can fully represent the underlying anatomy in difficult cases Caballo et al. [2020], Shim et al. [2022]. Forcing deterministic models to fit these labels can encourage overconfident boundary decisions and limit clinical reliability. Taken together, extreme data scarcity, severe class imbalance, and irreducible boundary ambiguity indicate that progress requires not merely a stronger discriminative architecture, but a probabilistic modeling paradigm that represents segmentation as a distribution rather than a point estimate.

Recent advances in generative modeling provide a principled alternative to deterministic segmentation in clinically ambiguous regimes. Diffusion-based methods have already shown that medical segmentation can be formulated as conditional distribution learning, where a mask is generated by iterative denoising rather than direct pointwise classification Wu et al. [2024], Rahman et al. [2023], Wolleb et al. [2022], Xing et al. [2025]. Across recent medical segmentation studies, this probabilistic perspective is increasingly valued for ambiguity-aware prediction in difficult boundary regions Wu et al. [2024], Rahman et al. [2023], Schmidt et al. [2023], Xing et al. [2025]. Latent-space modeling further improves tractability: latent diffusion work by Rombach et al. [2022] showed that high-resolution synthesis can be performed efficiently in compressed representations, establishing the computational rationale for latent generative segmentation pipelines Rombach et al. [2022]. However, dedicated BCT applications remain limited, particularly under high-resolution small-cohort settings.

Standard diffusion relies on stochastic differential dynamics with curved probability trajectories and often requires hundreds of iterative function evaluations, which is computationally expensive for high-resolution 3D data. Flow-Matching, introduced by Lipman et al. [2023], addresses this

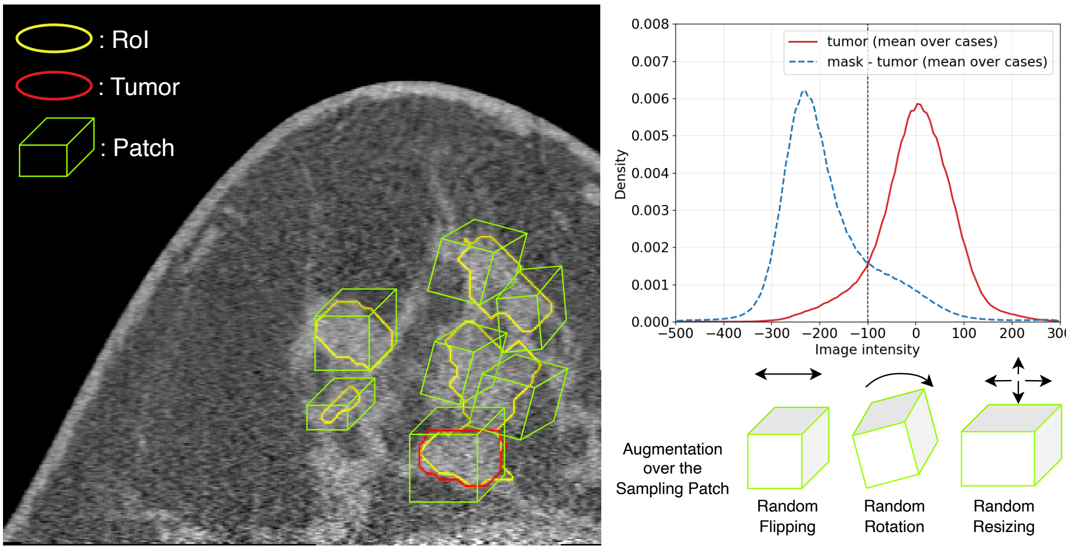

Figure 1. Data-side pipeline of the proposed framework: physics-informed candidate localization and ROI-centered 3D augmentation before model training/inference.

limitation by learning an ODE velocity field along linear conditional probability paths that transport samples from a simple source distribution to target masks Bogensperger et al. [2025]. As a result, simple ODE solvers such as Euler integration can reach high-quality segmentations in far fewer steps, in our case 25 inference steps, while reducing computational overhead relative to diffusion-style samplers Wang et al. [2024]. Empirical evidence supports this efficiency: SemFlow Wang et al. [2024] reports strong segmentation performance with 25-step inference relative to diffusion baselines requiring 200 steps, and MAISI-v2 Zhao et al. [2025] shows 33-fold acceleration for high-resolution 3D medical generation. For BCT specifically, this combination of computational efficiency and deterministic ODE inference is well aligned with small-data training and high-resolution deployment constraints. Building on these foundations, we design a physics-informed Flow-Matching framework tailored to the volumetric and attenuation characteristics of BCT.

The proposed framework integrates three coordinated design principles for BCT: physics-informed candidate localization from HU priors, ROI-centered 3D patch extraction with random affine augmentation, and Flow-Matching-based generative segmentation. First, HU thresholding and morphology-driven filtering restrict the search space to tumor-candidate regions, reducing background dominance before model training. Second, we sample fixed-size 3D patches ($64 \times 64 \times 64$ voxels) from HU-derived ROI bounding boxes and apply differentiable random transformations (anisotropic scaling, three-axis rotation, and random flipping), increasing effective training diversity while preserving volumetric anatomical context at tractable computational cost. Third, a lightweight Flow-UNet predicts the Flow-Matching velocity field on each extracted patch to generate segmentation distributions rather than a single deterministic mask. The specific contributions of this work are:

1. We propose a physics-informed generative segmentation framework that couples HU-based ROI extraction with Flow-Matching to improve tumor delineation under high-resolution, small-cohort BCT conditions.

2. We introduce an ROI-centered 3D patch augmentation strategy with random affine perturbations, which expands effective training variation from limited cohorts while preserving volumetric tumor context and enabling stable convergence in data-scarce regimes.

3. We adopt a detection-oriented evaluation under five-fold cross-validation and show that, relative to a deterministic UNet baseline, the generative formulation substantially reduces false positives per scan while maintaining comparable lesion-level detection sensitivity, reframing the benefit of generative BCT segmentation around deployment-relevant detection rather than overlap alone.

Together, these components maintain comparable tumor-level detection sensitivity to a deterministic UNet baseline while reducing false-positive burden, supporting the feasibility of generative BCT segmentation for screening-oriented decision support.

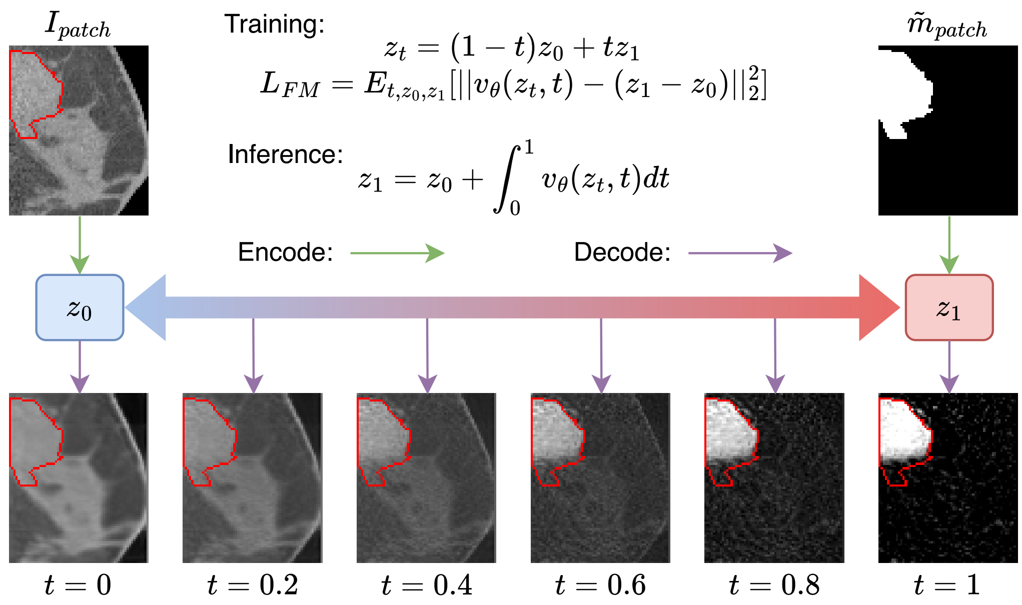

Figure 2. Model-side pipeline of the proposed latent Flow-Matching framework. Image and mask patches (I_{patch}, $\tilde{m}_{patch}$) are encoded into latent states (z_0, z_1). A velocity network v_θ is trained to map z_0 to z_1 via linear interpolation, and inference is performed through ODE integration. Decoded intermediate states from $t = 0$ to $t = 1$ visualize the continuous generation process, with the red outline indicating the ground-truth tumor boundary.

2 Methods

2.1 Overview

Figure 1 summarizes the data-side pipeline with three linked components: a physics-informed ROI proposal stage with HU-guided candidate localization (top-right density panel), ROI-centered spatial patch sampling around tumor candidates (left panel), and geometric 3D augmentation of sampled patches (bottom-right panel). Figure 2 summarizes the model-side pipeline. The source BCT image patch (I_{patch}) and the target mask patch ($\tilde{m}_{patch}$) are first encoded into latent representations z_0 and z_1. During training, the Flow-Matching objective learns a velocity field v_θ to transport z_0 toward z_1 along a linear interpolation path. At inference, the predicted mask latent is generated by integrating the velocity field over $t \in [0, 1]$. The decoded intermediate states illustrate a continuous transformation from the image domain at $t = 0$ to the binary segmentation mask at $t = 1$, with the ground-truth tumor boundary highlighted in red. The implementation uses four aligned volumetric channels for each sampled patch: BCT image intensity, breast mask, tumor-candidate mask, and tumor label.

2.2 Data representation and preprocessing

The clinical data were acquired with a dedicated BCT system (AB-CT[1] – Advanced Breast-CT GmbH, Erlangen, Germany) using photon-counting detector technology and isotropic high-resolution reconstruction. The analyzed cohort contains 35 patients, each with paired image and expert tumor annotation volumes. Tumor annotations are represented as binary voxel masks. In addition, a breast mask channel is used to define anatomically valid voxels. Data splitting follows patient-level 5-fold cross-validation, such that all patches from one patient remain within a single fold.

Input BCT intensities are clipped to a fixed HU window of $[-500, 300]$ and linearly rescaled to $[-1, 1]$ before training and inference:

$$I_{\text{norm}} = 2 \cdot \frac{\text{clip}(I, -500, 300) - (-500)}{300 - (-500)} - 1. \tag{1}$$

Only voxels inside the breast mask are treated as anatomically valid support for downstream learning and evaluation.

Because malignant tumors in BCT are typically hyperdense relative to surrounding dense parenchyma Weber et al. [2024], we adopt a Physics-Informed ROI Proposal to restrict the search

[1] `https://ab-ct.de/`

space. The top-right panel of Figure 1 shows the cohort-level intensity distributions for tumor and non-tumor breast tissue (mask − tumor), with an intersection near −100 HU; this crossing point is used as the candidate threshold. Let $I(x)$ denote the input volume and set $\tau_{\mathrm{HU}} = -100$. The Physics-Informed ROI Proposal applies HU thresholding first, followed by morphological erosion and then dilation before connected-component extraction. This erosion–dilation step suppresses thin, line-like glandular tissue candidates and retains more lesion-focused regions. Each connected component is then converted to an ROI box:

$$\mathcal{R}_k = \mathrm{BBox}\left(\mathrm{CC}_k\left(\mathrm{Dilate}\left(\mathrm{Erode}\left(\mathbf{1}[I(x) > \tau_{\mathrm{HU}}]\right)\right)\right)\right). \tag{2}$$

In code, each ROI is stored as a 3D integer bounding box $[\mathbf{s}_k, \mathbf{e}_k] \in \mathrm{Z}^{3\times 2}$, where $\mathbf{s}_k$ and $\mathbf{e}_k$ denote the start and end corner vectors of the k-th ROI, respectively. These ROIs reduce background dominance before learning and enforce tumor-focused sampling in high-resolution volumes.

2.3 ROI-centered 3D patch augmentation

Given a selected ROI $\mathcal{R}_k$, we sample a fixed-size 3D patch of $64 \times 64 \times 64$ voxels around a random center constrained inside the ROI box. As illustrated in the left panel of Figure 1, these sampled patches are spatially centered on HU-derived candidate regions and local tumor boundaries. Instead of first cropping and then applying image warps, the implementation perturbs the sampling grid directly and performs a single differentiable resampling call (`F.grid_sample`). This strategy preserves interpolation fidelity and keeps image/label correspondence exact across channels.

Let the canonical patch grid be $G_0 \in \mathrm{R}^{H\times D\times W\times 3}$ with axes centered at zero, where (H, D, W) are the patch dimensions (64, 64, 64 in our setting):

$$\mathcal{A}_h = \left\{-\frac{H}{2} + 0.5, \ldots, \frac{H}{2} - 0.5\right\}, \; \mathcal{A}_d = \left\{-\frac{D}{2} + 0.5, \ldots, \frac{D}{2} - 0.5\right\}, \; \mathcal{A}_w = \left\{-\frac{W}{2} + 0.5, \ldots, \frac{W}{2} - 0.5\right\}. \tag{3}$$

The meshgrid of $(\mathcal{A}_w, \mathcal{A}_d, \mathcal{A}_h)$ defines voxel-centered sampling coordinates before normalization. A random ROI-constrained center-offset vector $\mathbf{o}$ is then sampled uniformly:

$$\mathbf{o} \sim \mathcal{U}\left(\mathbf{s}_k + \mathbf{P}/2, \mathbf{e}_k - \mathbf{P}/2\right), \tag{4}$$

where $\mathbf{s}_k$ and $\mathbf{e}_k$ are the ROI start/end corners, $\mathbf{P}$ is patch size.

Training-time geometric perturbation follows the bottom-right panel of Figure 1: (i) random resizing implemented as anisotropic scaling $s_i \sim \mathcal{U}(0.7, 1.3)$ independently for each axis, (ii) random rotations within $\pm 30°$ around x, y, z axes via an axis-angle rotation matrix, and (iii) random flipping with probability 0.5. The transformed grid can be written as

$$G_{\mathrm{aug}} = R \cdot \mathrm{diag}(s_x, s_y, s_z) \cdot G_0/\mathbf{V} + \mathbf{o}. \tag{5}$$

where $R \in \mathrm{R}^{3\times 3}$ is the axis-angle rotation matrix, and (s_x, s_y, s_z) are the per-axis scaling factors sampled from $\mathcal{U}(0.7, 1.3)$. Image voxels are sampled with trilinear interpolation, while binary masks and labels use nearest-neighbor sampling, ensuring discrete supervision remains consistent.

At test time, augmentation is disabled. For each ROI box, patches are generated by deterministic sliding-window tiling with stride $32 \times 32 \times 32$. Along axis i, the number of windows is

$$n_i = \left\lceil \max\left(\frac{B_i - P_i}{S_i}, 0\right)\right\rceil + 1, \tag{6}$$

where B_i is ROI extent, P_i is patch size, and S_i is stride. Overlap regions are fused by voxel-wise averaging in reconstruction.

2.4 Latent Flow-Matching segmentation

2.4.1 Latent representation To make high-resolution 3D generation tractable, image and mask patches are compressed by a pre-trained MAISI AutoencoderKL Guo et al. [2025] implemented in the MONAI generative stack Cardoso et al. [2022]. The VAE uses three encoder stages with channel configuration [64, 128, 256], two residual blocks per stage, and GroupNorm-based normalization with 32 groups. With 8× spatial downsampling and 4 latent channels, each $64 \times 64 \times 64$ patch maps to a latent tensor of size $4 \times 8 \times 8 \times 8$. During Flow-Matching training, VAE parameters are frozen, and only the latent velocity predictor is optimized.

Before encoding tumor labels, uniform noise is injected:

$$\tilde{m} = m + \epsilon, \quad \epsilon \sim \mathcal{U}(-0.5, 0.5). \tag{7}$$

This perturbation is re-sampled at every training iteration. The goal is to avoid overly degenerate latent targets from strictly binary masks and to stabilize gradient flow for velocity regression in latent space.

2.4.2 Flow-Matching objective Our implementation uses **image-to-mask** Flow-Matching rather than noise-to-mask transport. Let

$$z_0 = \mathrm{Enc}(I_{\text{patch}}), \qquad z_1 = \mathrm{Enc}(\tilde{m}_{\text{patch}}), \tag{8}$$

where $\tilde{m}$ is the noise-perturbed mask, (I_{patch}, m) are the paired image and mask patches. Following Flow-Matching Lipman et al. [2023], we define a linear interpolation path

$$z_t = (1-t)z_0 + tz_1, \quad t \sim \mathcal{U}(0,1), \tag{9}$$

and train a velocity network $v_\theta(z_t, t)$ with target velocity $v^* = z_1 - z_0$. The training loss is

$$\mathcal{L}_{\text{FM}} = \mathrm{E}_{t,z_0,z_1}\left[\|v_\theta(z_t,t) - (z_1 - z_0)\|_2^2\right]. \tag{10}$$

Compared with standard noise-anchored transport ($z_0 \sim \mathcal{N}(0, I)$), this formulation starts from a structured image latent and learns the residual transformation toward the tumor-mask latent, consistent with the SemFlow design intuition Wang et al. [2024]. In practice, this shortens the effective transport path and improves convergence in small-data 3D settings. Time is sampled continuously from $\mathcal{U}(0,1)$ (no reweighting schedule), and the model predicts velocity directly rather than predicting the endpoint latent.

2.4.3 Network architecture and conditioning The velocity predictor is a lightweight 3D Flow-UNet with encoder-decoder topology and skip connections. With $n_{\text{blocks}} = 4$, encoder strides are $[1, 2, 1, 2]$, and channel progression is 64→128→128→256 (base width as 64). The input and output tensors are both $8 \times 8 \times 8$ latent volumes.

Time conditioning uses sinusoidal timestep embeddings of dimension 64, followed by an MLP to 256 channels. Each StackedBlock applies affine modulation to normalized activations:

$$h' = (\gamma(e) + 1) \cdot \mathrm{IN}(h) + \beta(e), \qquad [\gamma, \beta] = \mathrm{Linear}(e), \tag{11}$$

where e is the time embedding and IN denotes 3D instance normalization. Decoder upsampling uses transposed convolutions, concatenated with encoder skip features before block-wise refinement.

There is no separate image-concatenation conditioning branch. Conditioning is implicit in the interpolation state z_t: at $t = 0$, $z_t = z_0$ (pure image latent); at $t \to 1$, $z_t \approx z_1$ (near-mask latent). This time-varying state naturally encodes the image-to-mask trajectory while keeping model complexity low.

2.4.4 ODE-based inference At inference, generating the target mask is formulated as solving an initial value problem defined by the learned velocity field. Starting from the initial state $z_0 = \mathrm{Enc}(I_{\text{patch}})$, the predicted mask latent z_1 is obtained by integrating the velocity field over the normalized time interval $t \in [0, 1]$:

$$z_1 = z_0 + \int_0^1 v_\theta(z_t, t)dt. \tag{12}$$

To evaluate this integral numerically, we employ explicit Euler integration using 25 uniform discrete steps with a step size of $\Delta t = 1/25$:

$$z_{t+\Delta t} = z_t + \Delta t\, v_\theta(z_t, t). \tag{13}$$

The resulting final latent state z_1 is subsequently decoded back to the voxel space using the VAE decoder. A threshold of 0.3 is applied to the continuous voxel predictions to yield the binary tumor segmentation mask. Finally, overlapping local ROI patches are assembled into a full-volume prediction by averaging the overlapping regions.

Table 1. Main quantitative comparison between deterministic UNet and the proposed Flow-Matching framework over five-fold cross-validation. UNet predictions are from cross-validated `unet3D_b8` outputs; Flow-Matching predictions are from cross-validated latent Flow-UNet outputs. DSC is computed within the physics-informed tumor-candidate region. All metrics use prediction threshold 0.3, minimum lesion size 50 voxels, and IoU ≥ 0.5 matching. Detection sensitivity is the primary lesion-level endpoint (equivalently, one minus the lesion-level miss rate). False positives per scan quantify unmatched predicted connected components under the same criterion.

Method	Candidate-region DSC (mean±std)	Detection sensitivity (%)	False positives per scan
Deterministic UNet	0.40 ± 0.12	45.1 ± 20.7	3.0 ± 2.0
Flow-Matching (ours)	**0.44 ± 0.13**	44.7 ± 20.6	**1.5 ± 0.6**

2.5 Training and evaluation protocol
Optimization uses AdamW with learning rate 2×10^{-4}, $(\beta_1, \beta_2) = (0.9, 0.999)$, cosine learning-rate decay, 200 warmup steps, and gradient clipping with max norm 16. Training uses mixed precision (FP16), batch size 32, and 50 epochs per fold. The effective batch size is 32. Multi-GPU execution is handled through HuggingFace Accelerate[2].

Model quality is evaluated with voxel-level overlap and tumor-level detection metrics. At the lesion level, each connected component in the expert tumor annotation is treated as one annotated tumor after discarding fragments smaller than 50 voxels, which removes sparse boundary speckles that are not treated as separate clinical lesions. A tumor is considered detected if at least one predicted connected component of the same minimum size (obtained after binarizing predictions at 0.3) achieves intersection-over-union (IoU) ≥ 0.5 with that annotation. Tumor detection sensitivity is the fraction of annotated tumors that are detected; equivalently, it is one minus the lesion-level false-negative rate (miss rate). We treat sensitivity as the primary detection endpoint in this screening-oriented setting, because missed tumors are clinically more consequential than contour imprecision.

To quantify the complementary burden of over-detection under the same IoU matching rule, we additionally report false positives per scan: the mean number of predicted connected components (also after the 50-voxel size filter) that are not matched to any annotated tumor at IoU ≥ 0.5. This metric characterizes spurious lesion calls and downstream review load, but is interpreted as a secondary deployment-oriented endpoint relative to sensitivity. For voxel-level overlap, we report Dice Similarity Coefficient (DSC) within the physics-informed tumor-candidate region (cand), i.e., voxels flagged by the HU-based preprocessing pipeline in Section 2.2, rather than over the entire breast mask. All metrics are computed per fold and summarized as fold-wise mean ± standard deviation.

3 Results
3.1 Experimental setup
Results are reported on five-fold patient-level cross-validation with identical split logic for all methods. For each fold, models are trained from scratch and evaluated on the held-out patient subset. We summarize candidate-region DSC, tumor detection sensitivity (primary detection endpoint), and false positives per scan (secondary deployment endpoint), as defined in Section 2.5. Quantitative values are reported as fold-wise mean ± standard deviation. In addition, we report computational characteristics relevant to deployment, including the fixed 25-step Euler integration schedule for Flow-Matching inference and wall-clock prediction time per scan under the same hardware environment. This section presents observations only; interpretation is deferred to the Discussion.

3.2 Comparison with deterministic baseline
Table 1 summarizes the primary comparison between a deterministic UNet baseline and the proposed Flow-Matching framework. Flow-Matching improves mean candidate-region DSC from 0.40 to 0.44 and reduces false positives per scan from 3.0 to 1.5. Lesion-level detection sensitivity is comparable between methods (45.1% versus 44.7%), indicating that the main practical gain in this cohort is lower spurious-lesion burden rather than a large sensitivity increase.

At cohort level, Flow-Matching shows a modest candidate-region DSC gain and a clearer reduction in false positives per scan, while fold-wise detection sensitivity remains similar between methods. Per-fold values are stable without fold-specific outliers that invert the method ordering

[2] `https://huggingface.co/docs/accelerate/en/index`

Table 2. Ablation of ROI-centered 3D patch augmentation components on fold 0. Candidate-region DSC uses prediction threshold 0.3 and IoU $\geq$ 0.5 lesion matching (minimum lesion size 50 voxels), consistent with Table 1.

Augmentation variant	Candidate-region DSC	Detection sensitivity (%)	Convergence
No augmentation			No/unstable
Scale only			Yes
Scale + rotate			Yes
Full (scale + rotate + flip)	**0.544**	**75.0**	Yes

Table 3. Ablation of transport source formulation in Flow-Matching (five-fold cross-validation). Metrics are computed within the physics-informed tumor-candidate region using prediction threshold 0.3, minimum lesion size 50 voxels, and IoU $\geq$ 0.5 matching, consistent with Table 1.

Transport formulation	DSC (mean±std)	Detection sensitivity (%)	FP per scan
Noise-to-mask ($z_0 \sim \mathcal{N}(0, I)$, image latent concat.)	0.35 ± 0.11	23.8 ± 12.0	2.7 ± 1.0
Image-to-mask ($z_0 = \mathrm{Enc}(I)$, $z_1 = \mathrm{Enc}(m)$)	**0.44 ± 0.13**	**44.7 ± 20.6**	**1.5 ± 0.6**

on overlap metrics. Where significance testing is available (for example, paired tests across folds), the corresponding p-values should be reported alongside Table 1. Inference cost is additionally summarized by fixed-step ODE integration (25 Euler steps) and measured runtime per scan under the same execution environment for both methods.

3.3 Ablation study: augmentation efficacy

To quantify the role of ROI-centered 3D patch augmentation, we ablate geometric components while holding all other training settings fixed. Table 2 reports four variants on fold 0 only (augmentation ablation; main results in Table 1 remain five-fold): no augmentation, scaling only, scaling plus rotation, and full augmentation (scaling, rotation, and flipping). DSC is computed within the physics-informed tumor-candidate region (`cand_dice`), using the same evaluation protocol as Table 1. The full configuration yields the strongest overall balance between overlap and tumor detection.

3.4 Ablation study: transport formulation

To evaluate the impact of transport source design, we compare two Flow-Matching formulations under the same VAE, augmentation policy, training epochs, and inference procedure: (i) *noise-to-mask* transport with concatenated image-latent conditioning ($z_0 \sim \mathcal{N}(0, I)$, image latent concatenated to the Flow-UNet input), and (ii) *image-to-mask* transport, where $z_0 = \mathrm{Enc}(I)$ as in our main method.

Table 3 reports five-fold candidate-region DSC, tumor detection sensitivity, and false positives per scan under the same evaluation protocol as Table 1. Image-to-mask transport improves candidate-region DSC (0.44±0.13 versus 0.35±0.11), detection sensitivity (44.7% versus 23.8%), and false positives per scan (1.5 versus 2.7) relative to noise-to-mask with concatenated image-latent conditioning.

3.5 Ablation study: ODE integration steps

To characterize the inference cost–accuracy trade-off of Flow-Matching sampling, we sweep the number of Euler integration steps while holding the trained five-fold checkpoints, ROI proposal, prediction threshold, and evaluation protocol fixed. The sweep uses the same image-to-mask transport configuration as the main method and varies only the number of ODE steps at inference. Figure 3 reports the five-fold mean candidate-region DSC and tumor detection sensitivity for each step count. Performance improves rapidly from 1 to 10 steps, then approaches a plateau: candidate-region DSC increases only from 0.433 at 25 steps to 0.438 at 100 steps, while detection sensitivity remains unchanged at 44.7%. The 25-step operating point therefore captures nearly all measured performance while avoiding additional ODE evaluations.

3.6 Size-stratified detection analysis

To localize where detection succeeds and fails, we pool all annotated lesions of at least 50 voxels across the five folds (47 lesions) and stratify the lesion-level detection rate (best IoU $\geq$ 0.5 with any

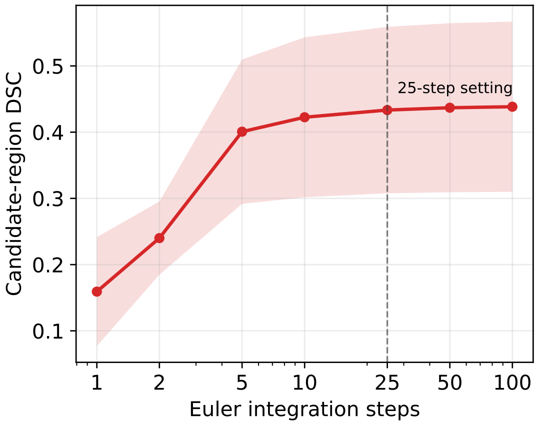

Figure 3. Effect of Flow-Matching Euler integration steps on segmentation performance. Candidate-region DSC and lesion-level detection sensitivity are plotted as functions of the number of ODE integration steps. Markers denote five-fold means, and shaded bands denote fold-wise standard deviation. All points use the same trained latent Flow-UNet checkpoints, prediction threshold 0.3, minimum lesion size 50 voxels, and IoU ≥ 0.5 lesion matching.

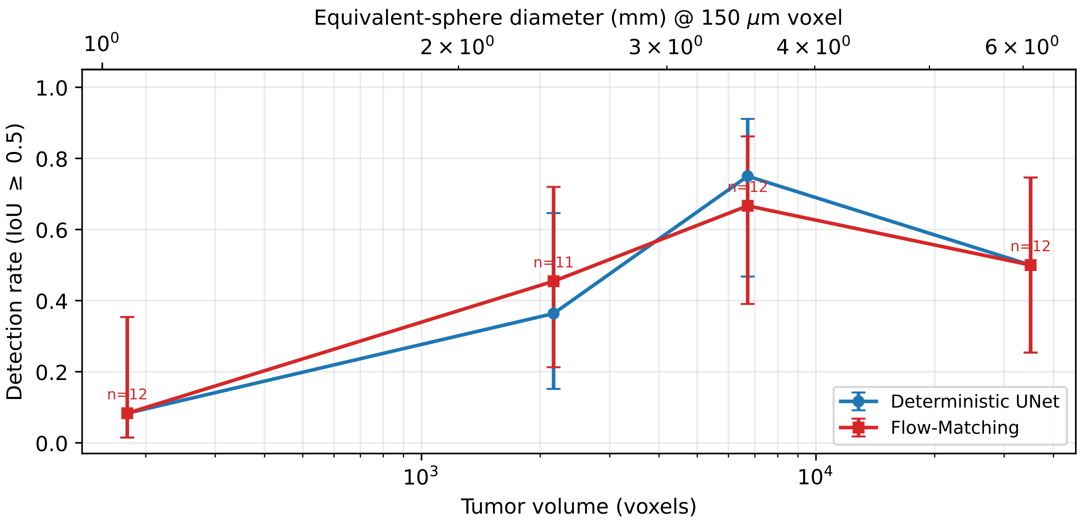

Figure 4. Lesion-level detection rate stratified by tumor size for Flow-Matching and the deterministic UNet baseline. Lesions (≥ 50 voxels) are pooled across the five folds and grouped into equal-count size quartiles; markers show the per-bin detection rate (best IoU ≥ 0.5) with Wilson 95% confidence intervals, and the top axis reports the equivalent-sphere diameter at 150 μm isotropic resolution. Detection rises monotonically with lesion size and is closely matched between the two methods, with misses concentrated in the smallest lesions.

predicted lesion, the same rule as Table 1) by lesion size. Figure 4 reports detection rate against tumor volume in equal-count size quartiles. Detection increases sharply with size for both methods: the smallest quartile (52–1131 voxels; median equivalent-sphere diameter ≈ 1 mm at 150 μm resolution) is detected only $\approx 8\%$ of the time, rising to 0.45 (Flow) and 0.36 (UNet) for 1.2k–3.5k-voxel lesions and to 0.67–0.75 for 4k–15k-voxel lesions; the largest quartile settles near 0.50 for both, as very large irregular masses are also harder to match under a strict IoU criterion. Across every size stratum the two methods track closely, and their pooled sensitivities are identical (42.6% each, consistent with Table 1). The residual misses are therefore concentrated in small, low-contrast lesions and are not specific to the generative formulation.

3.7 Detection operating characteristics

The single-threshold metrics in Table 1 capture one operating point; to characterize the full sensitivity–false-positive tradeoff we sweep the binarization threshold of each method's soft output and recompute pooled lesion-level sensitivity and mean false positives per scan under the same lesion definition. Figure 5(a) shows the resulting operating curves. Flow-Matching reaches its sensitivity plateau (≈ 0.43) at only ≈ 1.5 false positives per scan, whereas the deterministic UNet requires ≈ 3 false positives per scan to attain the same sensitivity; the Flow-Matching curve

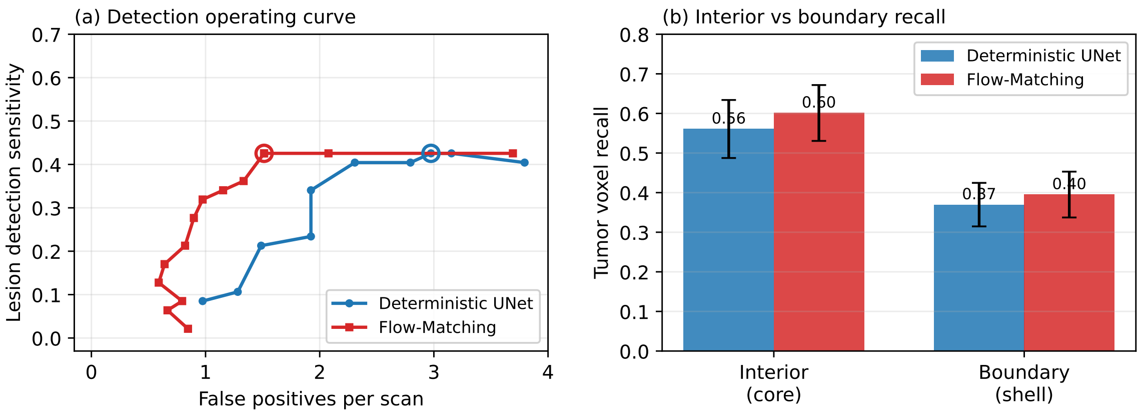

Figure 5. Detection characterization. (a) Operating curves (lesion sensitivity versus false positives per scan) obtained by sweeping the binarization threshold of each method, pooled over the cohort; circles mark the operating points used in Table 1, and at matched sensitivity Flow-Matching operates at roughly half the false-positive rate of the deterministic baseline. (b) Tumor-voxel recall in the interior core versus the boundary shell (2-voxel erosion), averaged over the 35 tumor-bearing cases (error bars: standard error of the mean); both methods recover the interior substantially better than the boundary, localizing the residual overlap error to partial-volume boundary voxels.

dominates (upper-left) throughout the clinically relevant low-false-positive regime. The reported operating points (Flow-Matching at threshold 0.3: 1.51 FP/scan; UNet at threshold 0.5: 2.97 FP/scan, both at 42.6% sensitivity) reproduce Table 1. Both curves saturate near 0.43 because roughly half of the lesions, predominantly the smallest, remain undetected at any threshold, consistent with the size analysis above. This confirms that the false-positive reduction in Table 1 reflects a genuine operating-curve advantage rather than a favorable threshold choice.

3.8 Boundary versus interior agreement

To identify where the residual voxel-level disagreement lies, we split each tumor into an eroded interior core and the complementary boundary shell (2-voxel erosion) and measure per-method tumor-voxel recall in each region over the 35 tumor-bearing cases. Figure 5(b) summarizes the result. Both methods recover the interior substantially better than the boundary (Flow-Matching 0.60 versus 0.40; UNet 0.56 versus 0.37), confirming that the overlap shortfall is dominated by the partial-volume boundary shell rather than by the lesion core. Flow-Matching is marginally higher than the UNet in both regions. This places the remaining error in physically ambiguous boundary voxels, where attenuation is mixed by partial-volume effects and a single correct contour is ill-defined.

3.9 Qualitative segmentation visualization

Figure 6 presents representative qualitative cases comparing deterministic UNet and Flow-Matching outputs on lesion-centered SBCT slices. The selected examples span three operating regimes observed in the quantitative analysis: a well-defined large lesion with high overlap for both methods, a boundary-disagreement case in which Flow-Matching better follows the annotated tumor extent, and a very small lesion close to the 50-voxel evaluation floor.

In the high-overlap case, both methods recover the main tumor support with similar contours, indicating that the latent generative formulation does not degrade segmentation when the lesion is visually conspicuous. In the boundary-disagreement case, the deterministic UNet produces a more conservative contour, whereas Flow-Matching captures a larger fraction of the annotated lesion boundary. For the near-threshold small lesion, neither method yields a matched predicted component under the lesion-level criterion, illustrating the failure mode underlying the size-dependent detection drop reported above.

4 Discussion

The comparison in Section 3.2 indicates that the proposed Flow-Matching framework provides a different performance profile from deterministic segmentation in small-cohort BCT. Although the absolute candidate-region DSC improvement is modest (0.40 to 0.44), false positives per scan decrease substantially (3.0 to 1.5), whereas lesion-level detection sensitivity remains comparable (45.1% versus 44.7%). This pattern suggests that overlap-based metrics alone do not fully characterize practical behavior in sparse-target volumetric screening tasks, and that

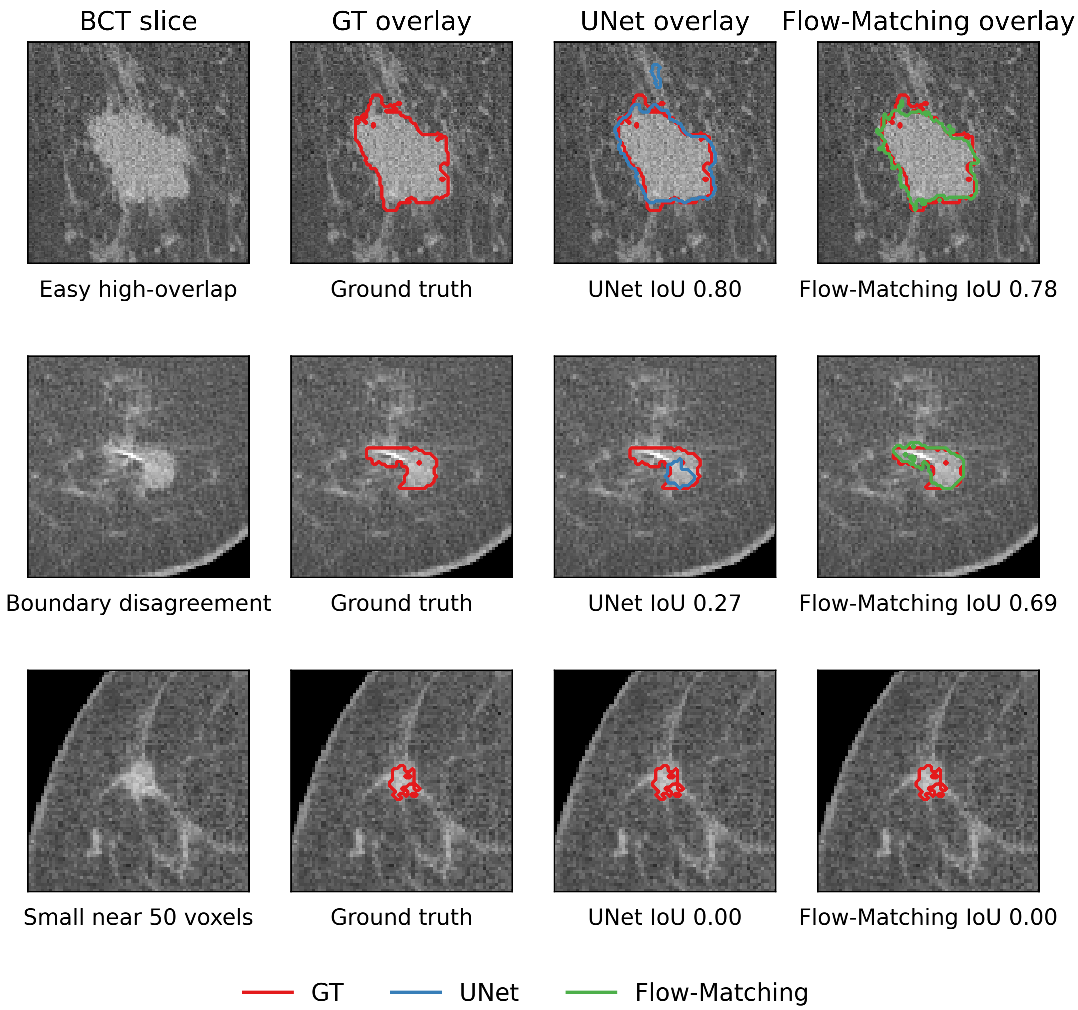

Figure 6. Representative qualitative segmentation outputs. Columns show the original SBCT slice, ground-truth overlay, deterministic UNet overlay, and Flow-Matching overlay. Red contours denote ground truth, blue contours denote UNet predictions, and green contours denote Flow-Matching predictions. Rows show a high-overlap case, a boundary-disagreement case, and a small lesion near the evaluation floor. Predictions are binarized at threshold 0.3 for Flow-Matching, and connected components smaller than 50 voxels are removed for display, matching the lesion-level evaluation protocol.

deployment-oriented false-positive burden should be interpreted alongside sensitivity. A plausible mechanism is that velocity-field learning in continuous time distributes supervision across interpolation states instead of forcing a single hard endpoint fit at every voxel. Under limited supervision, this can reduce brittle boundary decisions that arise when deterministic losses overfit sparse or noisy labels. This interpretation is consistent with the broader probabilistic-segmentation literature, where distribution learning is advantageous in anatomically ambiguous regions Schmidt et al. [2023], Wu et al. [2024], Rahman et al. [2023]. Relative to diffusion-style segmentation, the same probabilistic framing is retained while inference remains computationally tractable through fixed-step ODE integration Lipman et al. [2023], Wang et al. [2024], Zhao et al. [2025]. The step-count ablation further shows that the 25-step setting lies near the observed performance plateau, providing a practical accuracy–cost compromise for high-resolution 3D BCT deployment.

The transport ablation in Section 3.4 provides additional evidence for this design choice. In a standard noise-to-mask formulation, the model must reconstruct all structural information from an unstructured Gaussian source before reaching a plausible tumor configuration. In contrast, image-to-mask transport starts from an encoded image latent and learns the residual displacement toward the mask manifold, which shortens the effective path length in latent space. Under the same architecture and optimization budget, this structured-source formulation yields higher DSC than noise-to-mask in our cohort. The result is consistent with the intuition behind SemFlow-style conditional transport Wang et al. [2024]: when supervision is scarce and targets are small, reducing unnecessary transport complexity can improve optimization stability. The two formulations may also fail differently in practice. Noise-to-mask can generate anatomically implausible local structures, whereas image-to-mask may bias toward conservative updates (under-segmentation) when tumor evidence is weak. This tradeoff is important for interpreting method behavior beyond a single aggregate metric.

The augmentation ablation in Section 3.3 further suggests that ROI-centered 3D augmentation should be viewed as a training precondition rather than a minor optimization. In larger and more diverse datasets, augmentation often acts as a performance booster; in this setting, the no-augmentation variant can fail to converge stably, indicating insufficient geometric variation for robust optimization. This behavior is expected in BCT because tumor voxels occupy a tiny fraction of the full volume, and naive random sampling would overproduce background-only samples. By centering sampling on HU-derived candidates and perturbing local geometry through scale, rotation, and flipping, the pipeline preserves tumor context while materially increasing local appearance diversity. The design is also computationally aligned with high-resolution imaging: patch-level `F.grid_sample` transforms avoid expensive whole-volume augmentation passes while remaining differentiable and compatible with GPU training. Compared with standard volumetric augmentation practice in nnU-Net-style pipelines and related multi-planar segmentation frameworks Isensee et al. [2021], Pérez-García et al. [2021], Perslev et al. [2019], the present strategy is specifically adapted to the data-scale imbalance characteristic of emerging BCT cohorts.

An additional implementation detail is the latent-space label perturbation strategy. Binary labels encoded by a frozen VAE tend to occupy a narrow latent manifold, which can produce overly concentrated velocity targets and weak gradient diversity during training. Injecting bounded uniform perturbation before label encoding broadens the target distribution seen by the velocity regressor and acts as latent-space label smoothing. In practice, this stabilizes optimization without changing the semantic tumor support in voxel space. The uniform distribution is useful here because it preserves bounded perturbations around the binary support, avoiding occasional large excursions that can arise from unbounded Gaussian tails.

A related modeling tradeoff is introduced by freezing a pre-trained MAISI AutoencoderKL during task-specific Flow-Matching training. Freezing the VAE is beneficial in the small-cohort setting because it leverages a broad medical-imaging representation learned at larger scale and reduces the risk of overfitting the encoder-decoder to a narrow institutional distribution. However, the same choice fixes the compression ratio at $8\times$ and limits adaptation to BCT-specific small-target statistics, setting a representational floor for sub-clinical and boundary-scale structure that is a cost of representation rigidity accepted in exchange for training stability. A practical middle ground for future work may be selective adaptation (for example, decoder-side fine-tuning or low-rank adaptation) while preserving most of the foundation-model prior Zhao et al. [2025], Rombach et al. [2022], Pinaya et al. [2022].

These observations also motivate a metric-level discussion. Dice is a useful overlap index, but for small targets it is highly sensitive to minor boundary displacement. For example, mislabeling 10 boundary voxels in a 100-voxel tumor can reduce Dice by roughly 0.10, whereas the same absolute error in a 10,000-voxel structure produces a negligible change. In screening-oriented applications, the clinically dominant question is often tumor presence rather than contour perfection. We therefore report tumor-level detection sensitivity as the primary detection endpoint, which directly reflects missed annotated tumors (lesion-level false negatives). False positives per scan are reported as a complementary, deployment-oriented measure of spurious lesion calls under the same IoU matching rule; although false negatives are generally weighted more heavily than false positives in screening, excessive false positives can still limit clinical workflow acceptability. This hierarchy helps reconcile modest overlap gains with clearer improvements in detection-oriented utility.

From a computational-physics viewpoint, the choice of Flow-Matching over diffusion is also practical for high-resolution BCT deployment. Sliding-window inference over large 3D volumes requires evaluating many patches, and per-patch integration cost scales directly with the number of function evaluations (NFE). A diffusion-style sampler with 200 steps can be computationally prohibitive at volume level, whereas 25-step Euler integration reduces NFE by approximately an order of magnitude while preserving conditional generation quality in this setting Lipman et al. [2023], Wang et al. [2024], Zhao et al. [2025], Liu et al. [2023]. In addition, ODE-based integration is deterministic for a fixed input state, which supports reproducibility requirements in clinical and regulatory contexts.

Several limitations should be considered when interpreting these findings. First, the cohort size is limited and from a single-center acquisition setting, so external validity across scanners, protocols, and populations is not yet established Shim et al. [2022], Caballo et al. [2020]. Second, the comparative scope currently emphasizes a deterministic UNet baseline; broader benchmarking against additional strong baselines (for example, full nnU-Net pipelines and diffusion-segmentation variants) is still needed for a more exhaustive ranking. Third, the achieved DSC level remains below typical values reported in less challenging medical segmentation tasks, and the relative contributions of latent compression, annotation ambiguity, and data scarcity to this gap are not fully disentangled. Finally, this study focuses on technical feasibility and method characterization, not downstream diagnostic-outcome improvement; clinical utility therefore remains to be tested prospectively.

Future work will focus primarily on improving lesion-level detection, since detection sensitivity is the dominant limitation in this cohort. Promising directions include stronger tumor localization and lesion-level detection, more effective use of the limited annotated lesions, and loss designs that emphasize small-lesion recall. Reducing the latent-fidelity floor (lower-compression autoencoders, multi-scale latent pathways, or hybrid latent-pixel refinement) remains relevant mainly for resolving sub-clinical and boundary-scale structure. On the data side, larger multi-center BCT cohorts are needed to evaluate domain shift robustness and tumor-subtype behavior. On the benchmarking side, adding stronger deterministic and generative comparators will clarify which gains are specific to Flow-Matching and which reflect broader probabilistic modeling effects. For translational validation, prospective reader studies should quantify whether improved detection sensitivity (reduced missed tumors) and acceptable false-positive burden translate into measurable clinical workflow benefit. Finally, extending from binary tumor segmentation to richer multi-class tissue characterization may better leverage the quantitative HU information available in BCT and strengthen links between physical imaging signals and generative inference.

5 Conclusion

This study establishes that generative segmentation with Flow-Matching is a practical option for high-resolution BCT in the small-cohort regime. The central empirical message is a modest candidate-region overlap gain together with a marked reduction in false positives per scan, while lesion-level detection sensitivity remains comparable to a deterministic UNet baseline. The study also reveals that training behavior in this regime depends critically on how data variation is constructed: ROI-centered 3D geometric perturbation functions as a convergence condition under extreme class imbalance, not merely a minor performance enhancement. In addition, the residual performance gap is concentrated in small, low-contrast lesions, indicating that the dominant limitation in this cohort is lesion-level learning and detection rather than contour refinement. Taken together, these findings imply that progress in BCT segmentation is best framed as a joint modeling-and-physics problem, where the value lies in coupling probabilistic inference with modality-specific constraints so that algorithmic outputs remain consistent with known image ambiguity at tumor boundaries. More broadly, the results support a shift from pure overlap optimization toward evaluation strategies that emphasize missed-tumor rate (via detection sensitivity) and, secondarily, false-positive burden. At the same time, clinical translation requires cautious interpretation: larger multi-center cohorts, broader baseline comparisons, and prospective reader-impact studies are still needed before deployment-oriented claims can be made.

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